Crowbar Thyristor for the 3 kV / 3600 A Static-Switch Test Bench

Device Selection and Passive Self-Triggering Gate Circuit

Rev. 03 — 2026-09-11

Rev.	Date	Description
00	2026-09-09	First issue. Device selection, stress analysis, passive gate circuit.
01	2026-09-11	Added the simulated load-voltage waveform at breaker opening (Sec. 2) and the trigger-level and sizing chapter (Sec. 7): datasheet gate data of the 5STP 26N6500, choice of the negative load voltage that fires the crowbar, hold capacitor for the gate pulse, component-by-component sizing, computed gate-current waveforms, slow-opening case, failure modes. Corrected the $\mathrm{d}v/\mathrm{d}t$ estimate of Sec. 3.4 and the turn-on sequence of Sec. 6; thyristor data aligned with the datasheet.
02	2026-09-11	Added Sec. 3.2 on the role of the dump inductor L_c: commutation loop (Fig. 3), why the load inductance does not limit the turn-on $\mathrm{d}i/\mathrm{d}t$, trade-off with the MOV energy, requirements on L_c. Conclusions updated.
03	2026-09-11	Circuit figures (Figs. 1, 3, 4) enlarged for readability; battery symbol changed. No technical change.

Abstract

A thyristor crowbar with a series dump inductor and resistor extracts the energy of a 2 mH, 3600 A inductive load when a 3 kV IGBT static switch opens. This note quantifies the electrical stress on the crowbar thyristor, justifies the choice of a 6.5 kV symmetric phase-control thyristor (Hitachi Energy 5STP 26N6500), and shows that the intrinsic self-firing mechanisms of a thyristor cannot be relied upon: with the simulated rate of rise of the load voltage the device is in fact guaranteed *not* to fire, and without a gate circuit the varistor bank absorbs the whole load energy. A fully passive, floating anode-to-gate network turns the reversal of the load voltage into a regular gate pulse. Rev. 01 adds the simulated load-voltage waveform, derives from it the negative load voltage at which the crowbar fires, and gives a complete sizing of the network – blocking diode, coupling capacitor, TVS stack, hold capacitor, reset and gate-cathode elements – against the datasheet gate data, with computed gate-current waveforms, the slow-opening case and the failure modes. Rev. 02 adds a section on the role of the dump inductor L_c: it, and not the load inductance, sets the turn-on $\mathrm{d}i/\mathrm{d}t$, because the load inductance is not part of the loop in which the current transfers from the varistor to the thyristor.

1 Scope and circuit under study

The circuit is the static-switch test bench of Fig. 1: a DC source $V_s = 3000\,\mathrm{V}$ feeds an RL load ($R = 0.8\,\Omega$, $L = 2\,\mathrm{mH}$) through an IGBT breaker S (four 5SNA 1000G650300 in parallel).

The breaker is opened under load at $I_0 = 3600\,\mathrm{A}$; a varistor bank across the breaker clamps its voltage at $V_{\mathrm{MOV}} \approx 5.5\,\mathrm{kV}$. The load energy is extracted by the crowbar branch in parallel with the load: thyristor T, dump inductor $L_c = 12.5\,\mu\mathrm{H}$ and dump resistor $R_D = 0.42\,\Omega$, with an RC snubber $C_{sn} = 2\,\mu\mathrm{F}$, $R_{sn} = 64\,\Omega$ across the branch. The bench must perform of the order of 10^6 opening cycles.

The load voltage u_{load} is taken positive at the breaker-side terminal of the load. The thyristor anode is on the return rail, so that before the crowbar conducts

$$v_{AK} = -\,u_{\mathrm{load}}, \tag{1}$$

and the crowbar can conduct only when the load voltage is negative.

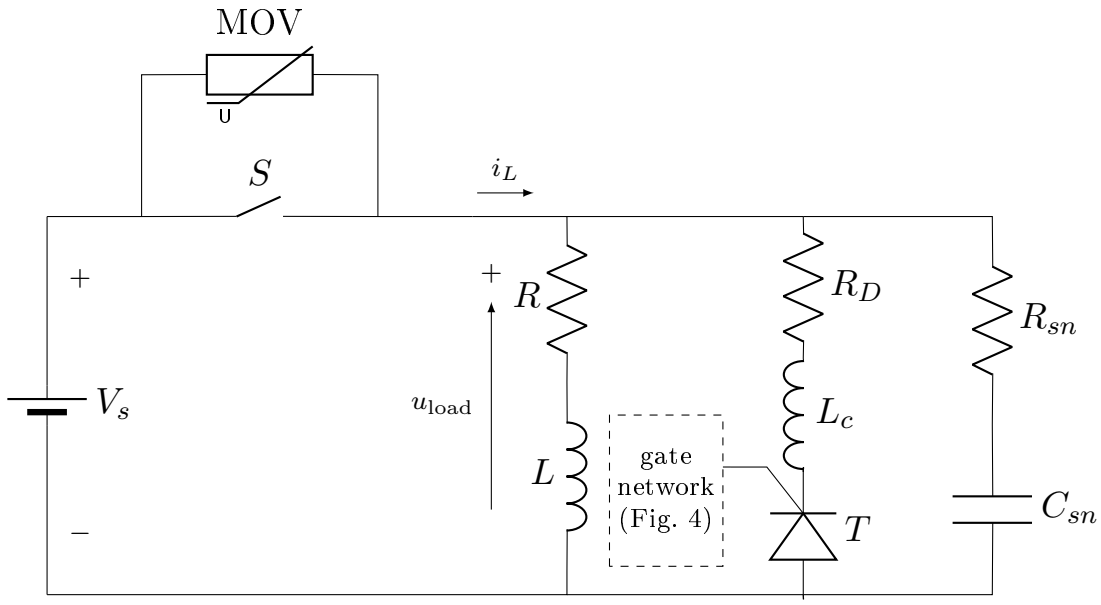


Figure 1: Static-switch test bench with dump-resistor crowbar. In service (S closed) the load voltage is $+V_s$ and the thyristor blocks in the *reverse* direction; forward voltage appears across T only while S opens.

Two facts shape the whole analysis:

1. In service the crowbar thyristor is *reverse* biased at $-V_s = -3\,\mathrm{kV}$. Forward voltage appears only when S opens and the load inductance reverses the load voltage, which is exactly when the crowbar must fire.

2. The forward voltage available to the crowbar is bounded by the MOV clamp of the breaker:

$$v_{AK,\max} = V_{\mathrm{MOV}} - V_s \approx 5.5\,\mathrm{kV} - 3.0\,\mathrm{kV} = 2.5\,\mathrm{kV} \tag{2}$$

as a plateau, with a first peak of about 2.8 kV (Sec. 2). Any firing threshold of the crowbar must lie well below this value, otherwise the MOV takes the load current and the energy.

2 Load voltage at breaker opening

2.1 Simulated waveform without crowbar

Fig. 2 shows the load voltage simulated for a breaker opening at $I_0 = 3600\,\mathrm{A}$ with the MOV bank as the only energy absorber (crowbar not firing). The features that matter for the trigger design are:

- The swing from +3 kV to about −2.5 kV is completed in roughly 4–5 µs; the maximum slope, around the zero crossing, is of the order of 1.5–2 kV/µs. This rate is set by the breaker (IGBT turn-off, its snubbers and the MOV loop), not by the crowbar branch.
- The first negative peak is about −2.8 kV: the varistor bank clamps at its highest voltage while it carries the full 3600 A, plus the drop on the loop inductance. The voltage then settles on the plateau $u_{\mathrm{load}} = V_s - V_{\mathrm{MOV}} \approx -2.5\,\mathrm{kV}$ and drifts to about −1.8 kV as the MOV current decreases.
- The MOV conducts for about 2.1 ms. With the MOV modelled as a constant back-voltage $V' = V_{\mathrm{MOV}} - V_s \approx 2.3\,\mathrm{kV}$ the load current is $i(t) = (I_0 + V'/R)\,e^{-t/\tau_L} - V'/R$ with $\tau_L = L/R = 2.5\,\mathrm{ms}$, which reaches zero at $t = \tau_L \ln(1 + I_0 R/V') \approx 2.0\,\mathrm{ms}$, in agreement with the simulation. The energy absorbed by the varistor bank in this mode is about 16.6 kJ per opening (the load energy $\frac{1}{2}LI_0^2 = 13\,\mathrm{kJ}$ plus the energy delivered by the source during the decay, minus the part dissipated in R).
- When the MOV current extinguishes, the load node rings with a positive peak of about 0.8 kV; for the thyristor this is reverse voltage and is harmless.

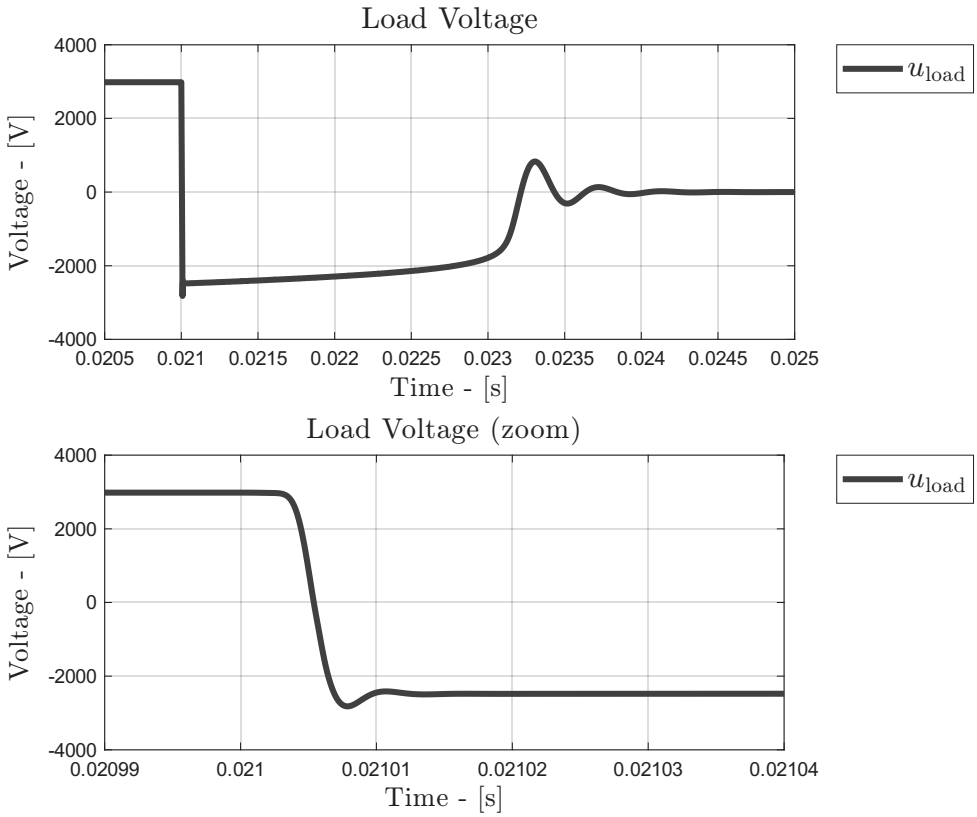


Figure 2: Simulated load voltage at breaker opening ($I_0 = 3600\,\mathrm{A}$), MOV clamp only, crowbar not firing. Top: full event; bottom: zoom on the 50 µs around the opening.

2.2 What the waveform means for the crowbar

A negative load voltage is an unambiguous "breaker open" signal. The circuit is DC and the load current is unidirectional. With the source connected, $u_{\mathrm{load}} = V_s > 0$; during the low-voltage pre-heating phase it is still positive. The only way for u_{load} to become negative is

that the path from the source is interrupted and the load inductance forces the current through whatever is left in parallel with the load – the MOV or the crowbar. No auxiliary logic is needed to decide when the crowbar must fire: $u_{\text{load}} < 0$, i.e. $v_{AK} > 0$ by (1), is both necessary and sufficient. The trigger network must therefore act only for $v_{AK} > 0$ and must be completely inert for $v_{AK} < 0$; the blocking diode D_1 of Sec. 6 does exactly this.

Without a gate circuit the crowbar does not fire. The critical $\mathrm{d}v/\mathrm{d}t$ of the 5STP 26N6500 is 3000 V/µs (exponential to $0.67\,V_{DRM}$, $T_{vj} = 125\,^{\circ}\mathrm{C}$) [1]. A ramp of 1.5–2 kV/µs reaching only 2.8 kV, at a junction temperature far below 125 °C, is inside the region where the device is guaranteed not to turn on. Forward breakover is even further away ($V_{BO} > V_{DRM} = 6.5\,\mathrm{kV}$). A crowbar thyristor without trigger circuit would thus behave exactly as in Fig. 2: it would block, and the varistor bank would absorb 16.6 kJ at every opening. The trigger network is not a refinement that makes the turn-on "controlled"; it is what makes the crowbar work at all.

Expected waveform with the crowbar. With the network of Sec. 6 the plateau at −2.5 kV lasts only for the gate delay plus the commutation of the current into L_c, about $3 + 18 = 21$ µs. The load voltage then jumps to $-(R_D\,i + v_T) \approx -1.5\,\mathrm{kV}$ and decays exponentially with $\tau = L/(R + R_D) = 1.64\,\mathrm{ms}$ to zero; the MOV conducts for 20 µs and absorbs about 180 J instead of 16.6 kJ. This is the waveform to be checked in the simulation once the crowbar and its network are added (Sec. 9).

3 Electrical stress on the crowbar thyristor

3.1 Conduction: charge, $\int i^2\,\mathrm{d}t$, energy

Once the load current has commutated into the crowbar branch, it decays through $R + R_D$ with

$$\tau = \frac{L}{R + R_D} = \frac{2\,\mathrm{mH}}{1.22\,\Omega} \approx 1.64\,\mathrm{ms}. \tag{3}$$

The conduction quantities follow directly:

$$Q = I_0\,\tau \approx 5.9\,\mathrm{C}, \tag{4}$$

$$\int i^2\,\mathrm{d}t = \frac{I_0^2\,\tau}{2} \approx 10.6 \times 10^3\,\mathrm{A^2s}, \tag{5}$$

$$E_L = \tfrac{1}{2} L I_0^2 \approx 13.0\,\mathrm{kJ}, \qquad E_{R_D} = E_L\,\frac{R_D}{R + R_D} \approx 4.5\,\mathrm{kJ}, \qquad E_R \approx 8.5\,\mathrm{kJ}. \tag{6}$$

The thyristor conduction energy, with the on-state model $v_T = V_{T0} + r_T\,i$ of the 5STP 26N6500 ($V_{T0} = 1.12\,\mathrm{V}$, $r_T = 0.29\,\mathrm{m\Omega}$), is

$$E_T = V_{T0}\,Q + r_T \int i^2\,\mathrm{d}t \approx 6.6 + 3.1 \approx 10\,\mathrm{J}. \tag{7}$$

Spread over the silicon of a 100 mm wafer (thermal capacity of the order of 10 J/K), this gives $\Delta T_j \lesssim 1\,\mathrm{K}$ per event. The crowbar thyristor is therefore *not* sized by current or by $\int i^2\mathrm{d}t$; it is sized by blocking voltage, turn-on $\mathrm{d}i/\mathrm{d}t$ and cycle count. The current reaches the holding level ($I_H = 125\,\mathrm{mA}$ at 25 °C) after $\tau \ln(I_0/I_H) \approx 17\,\mathrm{ms}$ and the device turns off naturally, without reverse recovery.

3.2 Role of the dump inductor L_c: the commutation loop

A natural objection is that the circuit already contains a large inductance, the 2 mH of the load, and that a further 12.5 µH in the crowbar branch can hardly limit anything. The objection

misses *where* the current changes. The load inductance is not part of the loop in which the current is transferred from the breaker to the crowbar, and on the time scale of that transfer it cannot change its current at all: with the load node at $-2.5\,\mathrm{kV}$ the load current decreases at $(V_{\mathrm{MOV}} - V_s + RI_0)/L \approx 2.7\,\mathrm{A/\mu s}$, i.e. by about 50 A out of 3600 A during the whole commutation. For the commutation, L is a constant current source I_0 (Fig. 3).

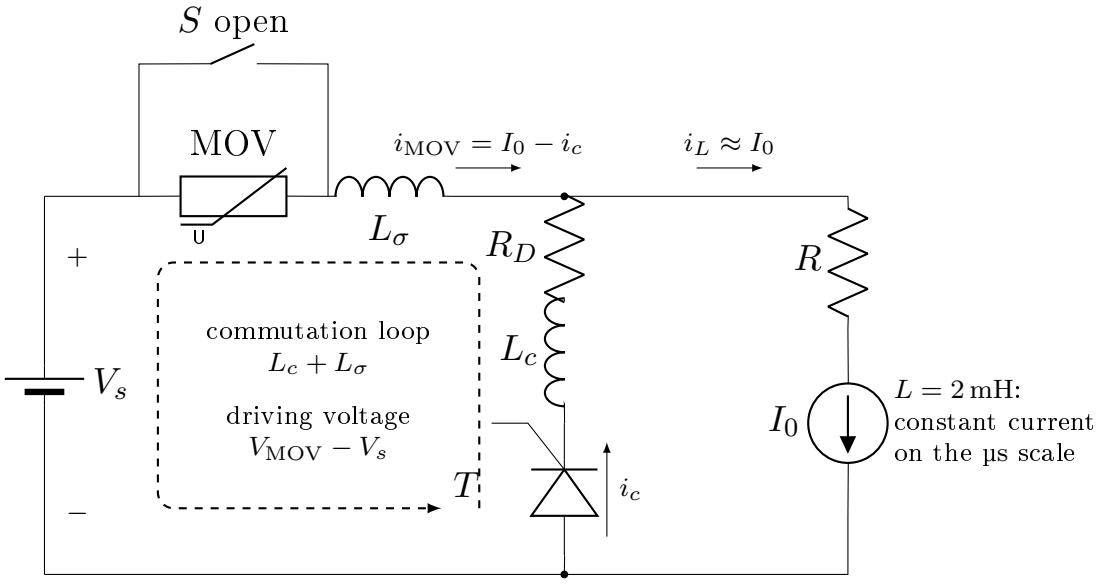


Figure 3: Commutation loop at crowbar firing. The load inductance carries a practically constant current I_0 during the transfer; the current rises in the thyristor as fast as it falls in the varistor, at a rate set by the driving voltage $V_{\mathrm{MOV}} - V_s$ and by the loop inductance $L_c + L_\sigma$, not by L.

At the instant the thyristor fires, I_0 flows through the breaker branch, i.e. through the varistor bank at V_{MOV}, and must be transferred to the crowbar branch. Kirchhoff's current law at the load node gives $i_{\mathrm{MOV}} = i_L - i_c \approx I_0 - i_c$: the rate at which the current rises in the thyristor is the rate at which it falls in the varistor, and it is set by the loop varistor–source–crowbar branch, whose inductance is L_c plus the stray inductance L_σ of the bus bars:

$$(L_c + L_\sigma)\,\frac{\mathrm{d}i_c}{\mathrm{d}t} = V_{\mathrm{MOV}} - V_s - R_D\,i_c - v_T. \tag{8}$$

Without L_c the loop would contain only L_σ, of the order of 1–2 µH for bus bars of a few metres: $\mathrm{d}i/\mathrm{d}t \approx 1.2$–$2.5\,\mathrm{kA/\mu s}$, six to twelve times the 50 Hz rating of the device, with the thyristor turning on at 2.5 kV on the small area around the amplifying gate. The situation is the same as the commutation between the switch and the freewheeling diode of a chopper: the inductive load keeps the total current constant, and the $\mathrm{d}i/\mathrm{d}t$ of the transfer between the two parallel paths is set by the loop inductance, not by the load.

The two inductors therefore play different roles:

- $L = 2\,\mathrm{mH}$ stores the energy (13 kJ), decides *how much* current has to be commutated (I_0) and, after the commutation, the decay $\tau = L/(R + R_D) = 1.64\,\mathrm{ms}$. On the microsecond scale it does not intervene.
- $L_c = 12.5\,\mu\mathrm{H}$ decides *how fast* the current transfers from the varistor to the thyristor, i.e. the turn-on $\mathrm{d}i/\mathrm{d}t$. After the commutation it is simply in series with L and no longer matters; its stored energy $\frac{1}{2}L_cI_0^2 = 81\,\mathrm{J}$ is negligible against E_L.

The price of L_c is that the varistor bank conducts during the whole transfer: $t_{\mathrm{comm}} = L_cI_0/(V_{\mathrm{MOV}} - V_s)$ and $E_{\mathrm{MOV,comm}} \approx \frac{1}{2}V_{\mathrm{MOV}}I_0t_{\mathrm{comm}}$ are both proportional to L_c (Sec. 3.3). L_c

is thus the trade-off between the $\mathrm{d}i/\mathrm{d}t$ of the thyristor and the energy per event in the MOV; the other knob is the clamp level, which enters (8) as the driving voltage. Consequences for the design of L_c: air-cored, so that it keeps its value at 3600 A (any saturation would make the $\mathrm{d}i/\mathrm{d}t$ collapse exactly when it is needed); insulated for the 2.8 kV that appear across it during the transfer; and with L_σ of the actual bus-bar geometry estimated, since it adds to L_c and is a 10–15 % correction in the right direction. The same inductor limits the $\mathrm{d}i/\mathrm{d}t$ of the abnormal re-closing case ($V_s/L_c = 240\,\mathrm{A/\mu s}$, Sec. 9) and the discharge current of the branch snubber at firing.

3.3 Commutation: $\mathrm{d}i/\mathrm{d}t$ and MOV energy

While S opens, the breaker voltage is held at V_{MOV} by the varistor bank; by (8), neglecting L_σ, $R_D i_c$ and v_T, the forward voltage (2) drives the load current into L_c:

$$\frac{\mathrm{d}i}{\mathrm{d}t} = \frac{V_{\mathrm{MOV}} - V_s}{L_c} = \frac{2.5\,\mathrm{kV}}{12.5\,\mu\mathrm{H}} = 200\,\mathrm{A/\mu s} \quad (224\,\mathrm{A/\mu s}\text{ at the }2.8\,\mathrm{kV}\text{ peak}), \tag{9}$$

$$t_{\mathrm{comm}} = \frac{L_c\, I_0}{V_{\mathrm{MOV}} - V_s} = 18\,\mu\mathrm{s}, \tag{10}$$

$$E_{\mathrm{MOV,comm}} \approx \tfrac{1}{2}\, V_{\mathrm{MOV}}\, I_0\, t_{\mathrm{comm}} \approx 180\,\mathrm{J}. \tag{11}$$

The MOV bank thus sees about 180 J per opening instead of the 16.6 kJ of Fig. 2, which is what makes the 10^6-cycle target credible. The turn-on $\mathrm{d}i/\mathrm{d}t$ is set by the MOV clamp level and L_c, not by the triggering method: it is the same whether the thyristor fires at 50 V or at 2 kV. The datasheet gives 200 A/µs repetitive at 50 Hz and 1000 A/µs at 1 Hz ($T_{vj} = 125\,^\circ\mathrm{C}$, $I_T = 3000\,\mathrm{A}$, $I_{GM} = 2\,\mathrm{A}$, $t_r = 0.5\,\mu\mathrm{s}$) [1]. The bench repeats the opening at a rate far below 50 Hz, at a low junction temperature and, with the network of Sec. 7, with a gate current of 5–7 A instead of 2 A: the 1 Hz figure is the relevant reference and the margin is comfortable. If the 50 Hz figure is to be respected literally, $L_c = 15$–$18\,\mu\mathrm{H}$ brings the commutation $\mathrm{d}i/\mathrm{d}t$ to 140–170 A/µs at the cost of $t_{\mathrm{comm}} = 22$–$26\,\mu\mathrm{s}$ and 215–260 J in the MOV.

The turn-on itself takes place at 2.5–2.8 kV: during the fall of the anode voltage (1–2 µs) and the subsequent spreading of the conducting area the device dissipates of the order of 1–2 J per event – less than the 10 J of conduction, but concentrated in the initially turned-on area. This is what the $\mathrm{d}i/\mathrm{d}t$ rating and the strong gate pulse of Sec. 7 protect.

3.4 $\mathrm{d}v/\mathrm{d}t$ at breaker opening

The rate of rise of the forward voltage is the one of Fig. 2:

$$\left.\frac{\mathrm{d}v_{AK}}{\mathrm{d}t}\right|_{\max} \approx 1.5\text{–}2\,\mathrm{kV/\mu s}, \tag{12}$$

so the thyristor voltage swings from −3 kV through zero to the 2.8 kV peak in about 4–5 µs. (Rev. 00 estimated an upper bound $I_0/C_{sn} = 1.8\,\mathrm{kV/\mu s}$ from the branch snubber alone; the figure is numerically similar but the reasoning was not correct, since the 64 Ω series resistor makes the branch snubber a high impedance at the switching edge. The rate is imposed by the breaker.) Compared with the critical 3000 V/µs of the device the margin is only a factor 1.5–2, which is the key observation of Section 5: the device will not self-fire, but the trigger network must make the gate region turn on before the intrinsic $\mathrm{d}v/\mathrm{d}t$ mechanism could, in a hotter device or with a faster breaker, act somewhere else on the wafer. If the breaker edge turns out faster than 3 kV/µs in the final design, additional capacitance at the load node (in RCD form, on the breaker side) is the correct remedy.

4 Device selection

4.1 Requirements

Table 1: Crowbar thyristor requirements and 5STP 26N6500 datasheet data [1].

Quantity	Requirement	5STP 26N6500	Note
Reverse blocking, continuous DC	3.0 kV	$V_{RRM} = 6.5\,\text{kV}$, $V_{RWM} = 4.34\,\text{kV}$	cosmic-ray FIT, see text
Forward blocking, transient	$\leq 2.8\,\text{kV}$	$V_{DRM} = 6.5\,\text{kV}$	bounded by MOV
Peak current	3600 A	$I_{TSM} = 65\,\text{kA}$, $I_{T(RMS)} = 4520\,\text{A}$	
$\int i^2\mathrm{d}t$ per event	$10.6 \times 10^3\,\text{A}^2\text{s}$	$21.1 \times 10^6\,\text{A}^2\text{s}$	10 ms surge
Turn-on $\mathrm{d}i/\mathrm{d}t$	200–224 A/µs	200 A/µs (50 Hz), 1000 A/µs (1 Hz)	Sec. 3.3
$\mathrm{d}v/\mathrm{d}t$ at turn-on	1.5–2 kV/µs	3000 V/µs critical	gate network required
Cycles	$\sim 10^6$	—	$\Delta T_j \lesssim 1\,\text{K}$
Gate trigger	—	$I_{GT} = 0.4\,\text{A}$, $V_{GT} = 2.6\,\text{V}$ (25 °C)	
Gate limits	—	$I_{FGM} = 10\,\text{A}$, $V_{RGM} = 10\,\text{V}$	
Gate delay	—	$t_{gd} = 3\,\mu\text{s}$ ($I_{GM} = 2\,\text{A}$, $t_r = 0.5\,\mu\text{s}$)	
Latching / holding	—	$I_L = 0.5\,\text{A}$, $I_H = 0.125\,\text{A}$ (25 °C)	

4.2 Voltage class: why 6.5 kV

The device blocks 3 kV DC continuously in the reverse direction for the whole life of the bench. For high-voltage silicon devices the dominant random failure mechanism under continuous DC blocking is cosmic-ray induced breakdown, whose rate rises very steeply with the ratio of applied to rated voltage. A 4.5 kV device would be operated at about 67 % of its rating, a 6.5 kV device at 46 % (69 % of its V_{RWM}). The FIT-versus-V_{DC} curves of the manufacturer must be consulted for the chosen part, but the 6.5 kV class is the natural choice; it also gives comfortable margin against the transients at breaker closing, when the device sees a $0 \rightarrow -3\,\text{kV}$ step.

The 5STP 26N6500 (Hitachi Energy, formerly ABB) is a symmetric 6.5 kV phase-control thyristor with interdigitated amplifying gate, $I_{T(RMS)} = 4520\,\text{A}$, $I_{TSM} = 65\,\text{kA}$ [1]. A Simscape model of this device with Lauritzen–Ma reverse recovery already exists, so the simulation work carries over. Its current margins in this application are very large (three orders of magnitude on $\int i^2\mathrm{d}t$); a smaller wafer of the same 6.5 kV family would be adequate thermally and would reduce clamping force and cost. The 100 mm device is retained here for the $\mathrm{d}i/\mathrm{d}t$ margin and because it is already modelled.

4.3 Why not an asymmetric pulse-power thyristor

Pulse-power thyristors (e.g. the Dynex range based on GTO structures, 3300 V and 4500 V, highly interdigitated gate, very high $\mathrm{d}i/\mathrm{d}t$) are marketed also for crowbar duty [6]. They do not fit here for two reasons: they are asymmetric and do not block the −3 kV seen in service (a series diode would be needed), and the $\mathrm{d}i/\mathrm{d}t$ of 200–224 A/µs imposed by L_c is within the reach of a phase-control device. A symmetric PCT is the simpler and more robust choice.

5 Self-firing mechanisms and why they must not be used as they are

5.1 $\mathrm{d}v/\mathrm{d}t$ firing

A rising forward voltage drives a displacement current $C_{J2}\,\mathrm{d}v/\mathrm{d}t$ through the blocking junction J_2. This current flows laterally in the p-base towards the cathode shorts; where the lateral voltage drop across the p-base under the n^+ emitter reaches $\approx 0.7\,\mathrm{V}$, the emitter injects and the thyristor turns on [3]. The location is wherever the p-base resistance is highest or the shorts are farthest, i.e. a random point of the cathode area, not the gate region. The datasheet "critical $\mathrm{d}v/\mathrm{d}t$" is the ramp the device is guaranteed *not* to fire on at $T_{j,\max}$; the actual firing ramp is unspecified, may be several times higher, and falls with temperature because leakage adds to the displacement current.

5.2 Forward breakover

When the forward voltage exceeds V_{BO} the avalanche multiplication at J_2 satisfies the regenerative condition $\alpha_{npn}\,\alpha_{pnp}\,M \to 1$ and the device turns on [3]. V_{BO} is not a datasheet parameter: it lies at roughly $1.1\text{–}1.3\times V_{DRM}$, varies with temperature and from device to device, and the turn-on initiates at the point of maximum field (junction termination or gate periphery). The undesired ignition of a thyristor occurs at a random, localized spot of the cathode junction and may destroy the device through the concentrated turn-on losses [4].

5.3 Consequences in this circuit

- Static self-firing on V_{BO} is not reachable: the forward voltage is limited to 2.8 kV by the MOV, far below the $V_{BO} > 6.5\,\mathrm{kV}$ of the chosen device.
- Dynamic self-firing is not reachable either, by guarantee: 1.5–2 kV/µs against a critical 3 kV/µs. The device blocks, and the MOV absorbs the full 16.6 kJ (Fig. 2).
- Should the intrinsic mechanism nevertheless act (hot device, faster breaker), the initially conducting area would be small and unrelated to the amplifying gate; the $\mathrm{d}i/\mathrm{d}t$ that follows (200 A/µs here) would be carried by that small area and the datasheet $\mathrm{d}i/\mathrm{d}t$ rating would not apply. Over 10^6 cycles this is not an acceptable operating mode.

The useful conclusion is to *pre-empt* these mechanisms: the same anode voltage that would cause an uncontrolled ignition is used to inject a proper gate current, so that turn-on always starts at the amplifying gate. This is the "protective firing" concept of HVDC valves, where a break-over diode (BOD) between anode and gate converts an overvoltage into a regular gate pulse [4, 5]. Here the concept is the *primary* trigger, and it must satisfy the gate-drive recommendations of the manufacturer [2].

6 Passive self-triggering gate network

6.1 Topology

All components are referenced to the cathode and float with it: no auxiliary supply, no fibre-optic link, no isolation transformer. With respect to Rev. 00 the hold capacitor C_h has been added and the reset resistor R_C now returns to the cathode instead of the gate rail, so that no reset current flows through the gate. The network works as follows.

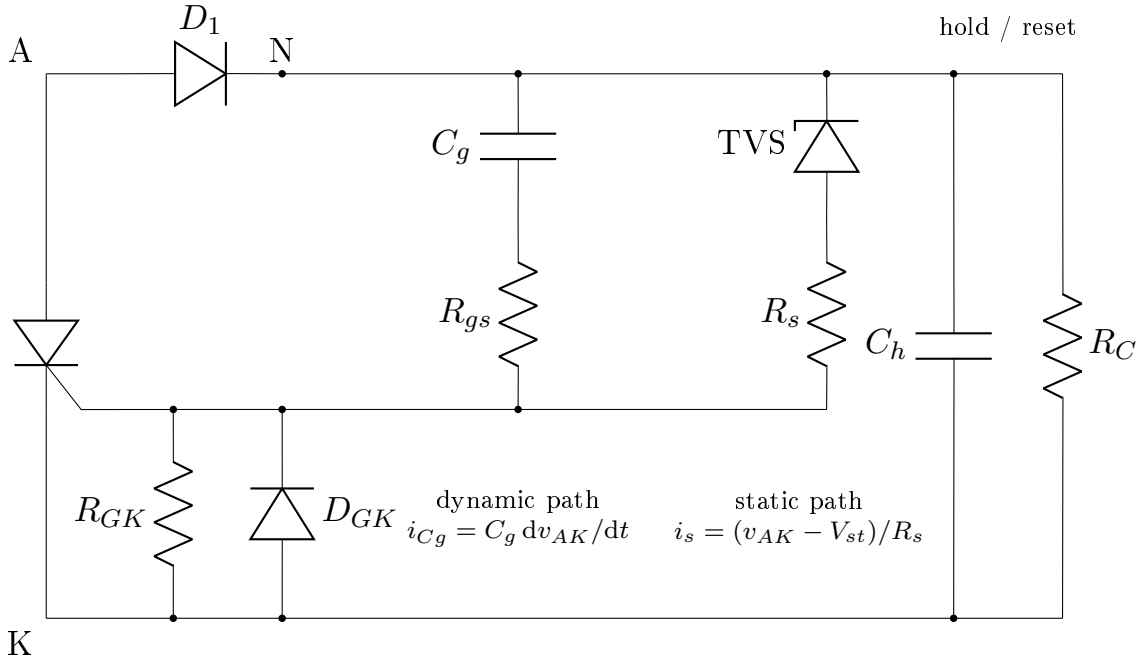


Figure 4: Floating anode-to-gate trigger network (Rev. 01). D_1 isolates the network during reverse blocking; C_g converts the forward $\mathrm{d}v/\mathrm{d}t$ into gate current; the TVS stack with R_s drives the gate as long as $v_{AK} > V_{st}$; C_h holds node N after the anode voltage collapses and keeps feeding the gate through the static path; R_C resets C_h; R_{GK} and D_{GK} protect the gate.

D_1 – **blocking (interlock) diode.** With $v_{AK} < 0$ (all of service life) D_1 is reverse biased and node N sits at cathode potential: no current reaches the gate. This is essential, because a positive gate current while the thyristor blocks reverse voltage forward-biases J_3 and J_2 and is amplified by the pnp transistor gain into an increased reverse leakage concentrated near the gate; manufacturers exclude this condition [2]. Without D_1, the coupling capacitor would inject $C_g\,\mathrm{d}v/\mathrm{d}t$ into the gate from the very start of the swing, i.e. while $v_{AK} = -3\,\mathrm{kV}$, and the static path would conduct in its forward direction at $-3\,\mathrm{kV}$. D_1 is what restricts the whole network to $u_{\mathrm{load}} < 0$ (Sec. 2.2).

Dynamic path – C_g, R_{gs}. As soon as v_{AK} crosses zero and rises, D_1 conducts and

$$i_{Cg} \approx C_g \frac{\mathrm{d}v_{AK}}{\mathrm{d}t} \tag{13}$$

flows into the gate through R_{gs} (valid for $R_{gs}C_g$ much shorter than the rise time). The gate is driven above the trigger level when the injected current exceeds I_{GT} plus what R_{GK} diverts:

$$\left.\frac{\mathrm{d}v}{\mathrm{d}t}\right|_{th} = \frac{I_{GT} + V_{GT}/R_{GK}}{C_g}. \tag{14}$$

The gate current starts within a few tens of volts of forward bias; the anode current starts t_{gd} later (3 µs at 2 A, less with a stronger pulse), when the anode voltage has already reached the MOV plateau. The dynamic path is the earliest trigger, but its current exists only while the voltage rises: on the plateau it is zero.

Static path – TVS stack and R_s. Once $v_{AK} > V_{st}$ the TVS stack conducts and drives the gate through R_s with $i_s = (v_{AK} - V_{st})/R_s$. This path has two roles: it fires the device if the forward voltage rises too slowly for the dynamic path (abnormal or mechanical opening), and

– the role that matters most in this bench – it *sustains* the gate current on the MOV plateau during the gate delay, when the dynamic path has already stopped. Since no forward voltage exists in service, V_{st} can be low; it must only exceed ringing and noise and remain far below (2). A stack of high-energy TVS diodes (5KP class) does the job; genuine BODs from HVDC gate-unit suppliers are the alternative.

Hold capacitor C_h. C_h charges through D_1 to the plateau voltage while the forward voltage rises. When the thyristor turns on and v_{AK} collapses, D_1 blocks and node N stays at 2.5–2.8 kV: the static path keeps feeding the gate from C_h with the time constant $R_s C_h$, well beyond the collapse of the anode voltage. This is what gives the gate pulse the 5–20 µs duration recommended by the manufacturer, which neither path can provide on its own once the anode voltage is gone.

R_C – reset. R_C discharges C_h (and, through R_{gs}, C_g) between two events, with $\tau_C = R_C C_h$ much longer than the event and much shorter than the cycle period; it also defines the potential of node N against the leakage of D_1 in service.

R_{GK}, D_{GK}. R_{GK} gives noise immunity and sets the threshold (14); D_{GK} (antiparallel) limits negative gate voltage to well below V_{RGM}, e.g. against the small displacement current through the D_1 junction capacitance at breaker closing, and protects the gate in the D_1-short fault (Sec. 7.8).

7 Trigger level and sizing of the network for the bench

7.1 Gate-drive requirements

The datasheet and the gate-drive application note of the manufacturer [1, 2] define what a good gate pulse is for this device: peak gate current $I_{GM} \approx$ 2–5 A, rate of rise $\mathrm{d}i_G/\mathrm{d}t \geq 2\,\mathrm{A/\mu s}$ ($t_r \leq 1\,\mu\mathrm{s}$), duration of the peak $t_p(I_{GM}) \approx$ 5–20 µs, back-porch current $I_{G,on} \geq 1.5\,I_{GT} = 0.6\,\mathrm{A}$ while the anode current builds up, and never more than $I_{FGM} = 10\,\mathrm{A}$. The $\mathrm{d}i/\mathrm{d}t$ ratings are specified with $I_{GM} = 2\,\mathrm{A}$, $t_r = 0.5\,\mu\mathrm{s}$. Translated to the passive network:

1. the gate current must exceed $I_{GT} + V_{GT}/R_{GK}$ within the first few hundred volts of forward bias and rise with $\mathrm{d}i_G/\mathrm{d}t \gg 2\,\mathrm{A/\mu s}$;
2. it must be 5–8 A on the plateau, during the gate delay, whatever the MOV tolerance;
3. it must remain above 2 A for several microseconds *after* the anode voltage has collapsed, and above 0.6 A for the 18 µs of the commutation;
4. it must never exceed 10 A, including the sum of both paths on the fastest plausible edge;
5. nothing must reach the gate for $v_{AK} < 0$, and the negative gate voltage must stay within −10 V at all times.

7.2 Which negative load voltage fires the crowbar

The question “at which negative load voltage should the crowbar fire” has two answers, because the two paths respond to different quantities.

Dynamic path: at the zero crossing. The dynamic path responds to the slope, not to the level. With the values of Table 3 the threshold (14) is

$$\left.\frac{\mathrm{d}v}{\mathrm{d}t}\right|_{th} = \frac{0.40\,\mathrm{A} + 2.6\,\mathrm{V}/10\,\Omega}{2.2\,\mathrm{nF}} \approx 300\,\mathrm{V/\mu s}, \tag{15}$$

five to seven times below the slope of Fig. 2. At the breaker opening the gate is therefore driven as soon as u_{load} crosses zero going negative: the "trigger level" of the dynamic path is $u_{\mathrm{load}} \approx 0^{-}$.

Static path: at $u_{\mathrm{load}} \approx -0.5\,\mathrm{kV}$, effective $-0.75\,\mathrm{kV}$. The static path starts conducting at $v_{AK} = V_{st}$ and reaches the gate trigger level when $(v_{AK} - V_{st})/(R_s + r_{TVS}) = I_{GT} + V_{GT}/R_{GK}$, i.e. at

$$V_{\mathrm{fire}} = V_{st} + (I_{GT} + V_{GT}/R_{GK})(R_s + r_{TVS}) \approx 520\,\mathrm{V} + 0.66\,\mathrm{A} \times 336\,\Omega \approx 0.75\,\mathrm{kV} \tag{16}$$

at 25 °C, lower on a warm device (I_{GT} falls with temperature). This is the level that governs a slow opening (Sec. 7.6); at the IGBT opening it is reached 0.4 µs after the zero crossing and merely adds to the dynamic pulse.

Choice of V_{st}. The constraints on V_{st} are weak, which is the strength of this application: there is no forward voltage in service, so the lower bound is only immunity to ringing (the 0.8 kV ringing of Fig. 2 is reverse and disappears with the crowbar working; residual ringing after the natural turn-off at the end of conduction carries $\frac{1}{2}LI_H^2 \sim$ µJ). The upper bound is the plateau at the lowest MOV tolerance, about 2.0 kV for a −10 % bank, with the requirement that the sustaining current on that plateau is still 4–5 A. A stack of three 5KP150A ($V_{BR} = 167$–$185\,\mathrm{V}$ each) gives $V_{st} = 500$–$555\,\mathrm{V}$: the effective firing level (16) is 2.7 times below the lowest plateau and 3.7 times below the nominal one. Two devices ($V_{st} \approx 350\,\mathrm{V}$) would be equally acceptable and would fire a slow opening earlier; more than four ($V_{st} > 0.7\,\mathrm{kV}$) would start to erode the sustaining current at low MOV tolerance and are not recommended.

In summary: **the crowbar fires at $u_{\mathrm{load}} \approx 0$ on the IGBT opening, and at $u_{\mathrm{load}} \approx -0.75\,\mathrm{kV}$ (design window -0.5–$-1.0\,\mathrm{kV}$) on any slow reversal of the load voltage; both are far below the -2.0–$-2.8\,\mathrm{kV}$ imposed by the MOV.**

7.3 Sequence of events

Table 2 gives the sequence for the nominal case (slope 1.8 kV/µs, $t_{gd} = 2.5$ µs assumed for the 5–7 A gate pulse, values of Table 3); the times are counted from the zero crossing of v_{AK} and come from the model of Sec. 7.5.

7.4 Component sizing

Coupling capacitor C_g. The peak dynamic current must stay below I_{FGM} on the fastest edge the breaker can produce; taking the critical $\mathrm{d}v/\mathrm{d}t$ of the device itself as the upper bound of the design space,

$$C_g \le \frac{I_{FGM}}{(\mathrm{d}v/\mathrm{d}t)_{\max}} = \frac{10\,\mathrm{A}}{3\,\mathrm{kV/\mu s}} = 3.3\,\mathrm{nF}. \tag{17}$$

$C_g = 2.2\,\mathrm{nF}$ gives 4 A at 1.8 kV/µs and 6.6 A at 3 kV/µs, with the threshold of 300 V/µs computed above. The capacitor sees the full forward swing (2.8 kV) at every event and 3 kV DC across the string D_1–C_g in service; a rating of 4 kV DC (HV ceramic disc or film) is required. It is realised as two 4.7 nF, 4 kV capacitors in series, so that a single shorted capacitor leaves 4.7 nF (8.5 A at 1.8 kV/µs) instead of a direct connection between node N and the gate (Sec. 7.8).

Table 2: Sequence of events at breaker opening, nominal case.

t	Event
$-2.6\,\mu s$	IGBTs turn off; u_{load} starts falling from $+3\,kV$; D_1 reverse biased, nothing reaches the gate.
0	$v_{AK} = 0$. D_1 conducts; C_g injects $\approx 4\,A$ within $50\,ns$ ($di_G/dt \sim 100\,A/\mu s$); C_h starts charging through D_1 ($\approx 40\,A$ through D_1).
$0.3\,\mu s$	$v_{AK} = V_{st}$: TVS stack conducts, i_s rises with v_{AK}; total gate current exceeds $5\,A$ at $0.7\,\mu s$.
$2.4\,\mu s$	v_{AK} peaks at $\approx 2.8\,kV$ (MOV at full current); dynamic path current ends; static path $\approx 6.8\,A$; total gate current peak $7.1\,A$ at $1.8\,\mu s$, $v_{GK} \approx 6\,V$.
2.5–$3\,\mu s$	Anode current starts at the amplifying gate (t_{gd}); rises at 200–$224\,A/\mu s$ set by L_c; v_{AK} collapses in 1–$2\,\mu s$; D_1 blocks, node N held at $\approx 2.8\,kV$ by C_h.
4–$20\,\mu s$	C_h discharges through TVS and R_s into the gate: $i_G > 2\,A$ until $10\,\mu s$, $> 0.6\,A$ until $18\,\mu s$; commutation of I_0 into L_c completes at $\approx 21\,\mu s$, MOV current $\to 0$.
$21\,\mu s$–$17\,ms$	Conduction; $u_{\text{load}} \approx -(R_D i + v_T)$ decays with $\tau = 1.64\,ms$; C_h and C_g reset through R_C ($\tau_C = 0.5\,ms$).
$\approx 17\,ms$	$i < I_H$: natural turn-off, $u_{\text{load}} \to 0$. Network ready for the next cycle; the next closing applies $v_{AK} = -3\,kV$ with D_1 blocking.

Series resistor R_{gs}. It damps the loop formed by C_g and the gate-lead inductance: for $L_{lead} = 0.2$–$0.3\,\mu H$, $2\sqrt{L_{lead}/C_g} = 19$–$23\,\Omega$, hence $R_{gs} = 22\,\Omega$; the time constant $R_{gs}C_g = 48\,ns$ does not delay the pulse. Pulse power $4\,A^2 \times 22\,\Omega = 350\,W$ for $2\,\mu s$: any 1–2 W pulse-rated resistor.

TVS stack. Three 5KP150A in series, unidirectional, cathodes towards node N (reverse breakdown from N towards the gate): $V_{st} = 500$–$555\,V$, dynamic resistance $r_{TVS} \approx 6\,\Omega$ for the stack. Peak current $6.8\,A$ at the $2.8\,kV$ peak; energy $V_{st}\,Q_s \approx 520\,V \times 55\,\mu C \approx 30\,mJ$ per event, $10\,mJ$ per device – negligible against the $5\,kW$ / $10/1000\,\mu s$ pulse rating of the 5KP class [7]. In service the stack is not stressed (D_1 blocks); in the reverse-gate direction it is protected by D_{GK}.

Series resistor R_s. The plateau current must not exceed I_{FGM} together with the dynamic contribution on the fastest edge, and must remain 4–5 A at the lowest plateau:

$$R_s \geq \frac{v_{AK,pk} - V_{st}}{I_{FGM} - i_{Cg}} \approx 300\,\Omega; \qquad R_s = 330\,\Omega: \quad i_s = \frac{2500\,V - 520\,V}{336\,\Omega} = 5.9\,A, \tag{18}$$

$6.8\,A$ at the $2.8\,kV$ peak and $4.4\,A$ at a $2.0\,kV$ plateau. The resistor carries $2.3\,kV$ during the pulse and about $75\,mJ$ per event (including the discharge of C_h): a chain of four $82\,\Omega$, 2–3 W metal-oxide resistors keeps the pulse voltage per resistor below $600\,V$; the average power at one event per second is $75\,mW$.

Hold capacitor C_h. After the collapse of v_{AK} the static-path current decays from $(V_{pl} - V_{st})/R_s$ with $\tau_h = (R_s + r_{TVS})C_h$. The requirement $i_G > 2\,A$ for at least $5\,\mu s$ after the collapse gives

$$\tau_h \ln \frac{5.9\,A}{2\,A} \geq 5\,\mu s \;\Rightarrow\; \tau_h \geq 4.6\,\mu s \;\Rightarrow\; C_h \geq 14\,nF. \tag{19}$$

$C_h = 22\,nF$ gives $\tau_h = 7.4\,\mu s$: $i_G > 2\,A$ for $8\,\mu s$ and $> 0.6\,A$ for $17\,\mu s$ after the collapse, i.e. a back-porch current over practically the whole commutation. The price is the charging current

through D_1, $(C_h+C_g)\,\mathrm{d}v/\mathrm{d}t \approx 44\,\mathrm{A}$ at $1.8\,\mathrm{kV/\mu s}$ (72 A at $3\,\mathrm{kV/\mu s}$) for about 2 µs, and the stored energy $\frac{1}{2}C_h(2.8\,\mathrm{kV})^2 = 86\,\mathrm{mJ}$ per event, dissipated in R_s, the TVS and R_C. Rating 4 kV DC (film, pulse-capable); 33 nF extends the back-porch to the full 18 µs at the cost of 66 A through D_1 and is the upper practical value.

Reset resistor R_C. Between N and K. It must discharge C_h well before the next closing (the crowbar conducts for 17 ms, the cycle period is much longer) and keep node N close to cathode potential against the leakage of D_1: with $I_{leak} \le 50\,\mu\mathrm{A}$ and $R_C = 22\,\mathrm{k\Omega}$ the offset is 1.1 V, $\tau_C = R_C C_h = 0.48\,\mathrm{ms}$, and the residual energy $\frac{1}{2}C_h V_{st}^2 \approx 3$–$5\,\mathrm{mJ}$ per event is dissipated there. The resistor sees 2.8 kV during the event and 3 kV continuously in the D_1-short fault (410 W): a chain of four 5.6 kΩ flame-proof (fusible) resistors, so that this fault opens the chain instead of starting a fire.

Blocking diode D_1. Requirements: reverse voltage 3.0 kV DC continuously, plus the closing overshoot of the breaker, and 2.8 kV for 0.5 ms after each event (node N held by C_h while the thyristor conducts); forward pulse 44–72 A for 2 µs followed by 6–7 A for a few microseconds, negligible I^2t and average current; *fast* recovery, because at the collapse of v_{AK} the recovery charge of D_1 is drawn from C_h into the thyristor anode – a standard-recovery diode with $Q_{rr} =$ 20–50 µC would cancel most of the 62 µC stored in C_h, whereas a fast diode ($Q_{rr} \lesssim 2\,\mu\mathrm{C}$, $t_{rr} \le 200\,\mathrm{ns}$) costs less than 3 %. A string of five 1.6 kV (or six 1.2 kV) fast-recovery diodes of the 10–30 A class, rated 8 kV in total, gives a 37 % DC utilisation and makes the D_1-short fault remote. Static sharing resistors across each diode, sized for at least five times the leakage spread of the chosen part (typically 1–4.7 MΩ, 0.5–1 W, ≥ 1 kV rated), dynamic sharing capacitors (100 pF, 2 kV) optional; the string capacitance of a few tens of picofarads passes only 0.05–0.1 A into node N at breaker closing, absorbed by C_h. Monolithic HV fast-recovery diodes (8–10 kV, 100 ns class) are an alternative if their non-repetitive surge rating covers the 2 µs charging pulse.

R_{GK}, D_{GK}. $R_{GK} = 10\,\Omega$, 2 W: diverts $V_{GT}/R_{GK} = 0.26\,\mathrm{A}$ at the trigger level and about 0.6 A at 6 V on the plateau, sets the dynamic threshold and shunts the displacement current of the D_1 capacitance. D_{GK}: fast or Schottky diode, 10 A class, antiparallel to the gate; it clamps $v_{GK} \ge -0.7\,\mathrm{V}$ ($V_{RGM} = 10\,\mathrm{V}$) and carries the fault current of the D_1-short case (≈ 9 A) until R_s opens.

Layout. The network is a small board mounted at the cathode side of the press-pack, floating at the load-node potential (+3 kV in service, −2.8 kV at the event, referred to the return rail): creepage and clearance for 6 kV between the D_1/node-N side and the cathode side, and system-level insulation of the board. D_1 connects to the anode-side bus bar with a short lead; the gate lead and its cathode reference are twisted and as short as the device allows, and R_{gs}, C_g, R_{GK}, D_{GK} are placed at the gate-lead end of the board to keep the gate loop small. A small current transformer on the gate lead is worth providing as a test point for commissioning and periodic checks (Sec. 7.8).

7.5 Computed gate-current waveforms

Fig. 5 shows the response of the network of Table 3 to the anode voltage of Fig. 2, computed with a simple time-domain model (`gate_network_sim.py`): $v_{AK}(t)$ is *forced* – raised-cosine swing from −3 kV to the 2.8 kV peak in 5 µs (slope $1.8\,\mathrm{kV/\mu s}$), decay to the 2.5 kV plateau, turn-on 2.5 µs after the zero crossing with a 1.5 µs voltage fall – and the network is integrated with D_1 as a 5 V / 1 Ω string, an ideal TVS knee with 6 Ω slope, and the gate-cathode characteristic linearised as $v_{GK} = 1.2\,\mathrm{V} + 0.7\,\Omega\, i_G$. The model replaces nothing of the Simscape verification; it only shows the timing of the two paths.

Table 3: Design values of the gate network (5STP 26N6500).

Element	Value	Design check
D_1	$5 \times 1.6\,\mathrm{kV}$ fast diodes, 8 kV string	3 kV DC reverse; 44–72 A / 2 µs forward; $Q_{rr} \lesssim 2\,\mu\mathrm{C}$; sharing R (and C) per diode
C_g	2.2 nF ($2 \times 4.7\,\mathrm{nF}$, 4 kV in series)	4 A at 1.8 kV/µs, 6.6 A at 3 kV/µs $< I_{FGM}$; threshold 300 V/µs
R_{gs}	22 Ω, pulse rated	$2\sqrt{L_{lead}/C_g} \approx 20\,\Omega$; $R_{gs}C_g = 48\,\mathrm{ns}$
TVS	$3 \times$ 5KP150A, $V_{st} = 500$–555 V	$V_{\mathrm{fire}} \approx 0.75\,\mathrm{kV}$, 2.7 × below the lowest plateau; 10 mJ/device/event
R_s	330 Ω ($4 \times 82\,\Omega$, 2–3 W MOX)	5.9 A at 2.5 kV, 6.8 A at 2.8 kV, 4.4 A at 2.0 kV; 75 mJ/event
C_h	22 nF, 4 kV DC film	$\tau_h = 7.4\,\mu\mathrm{s}$; $i_G > 2\,\mathrm{A}$ for 8 µs, $> 0.6\,\mathrm{A}$ for 17 µs after collapse; 86 mJ/event
R_C	22 kΩ ($4 \times 5.6\,\mathrm{k}\Omega$, fusible)	$\tau_C = 0.48\,\mathrm{ms}$; 1.1 V offset at 50 µA leakage; opens on D_1 short
R_{GK}	10 Ω, 2 W	0.26 A at V_{GT}; sets (14)
D_{GK}	fast/Schottky, 10 A class	$v_{GK} \geq -0.7\,\mathrm{V} \gg -V_{RGM}$

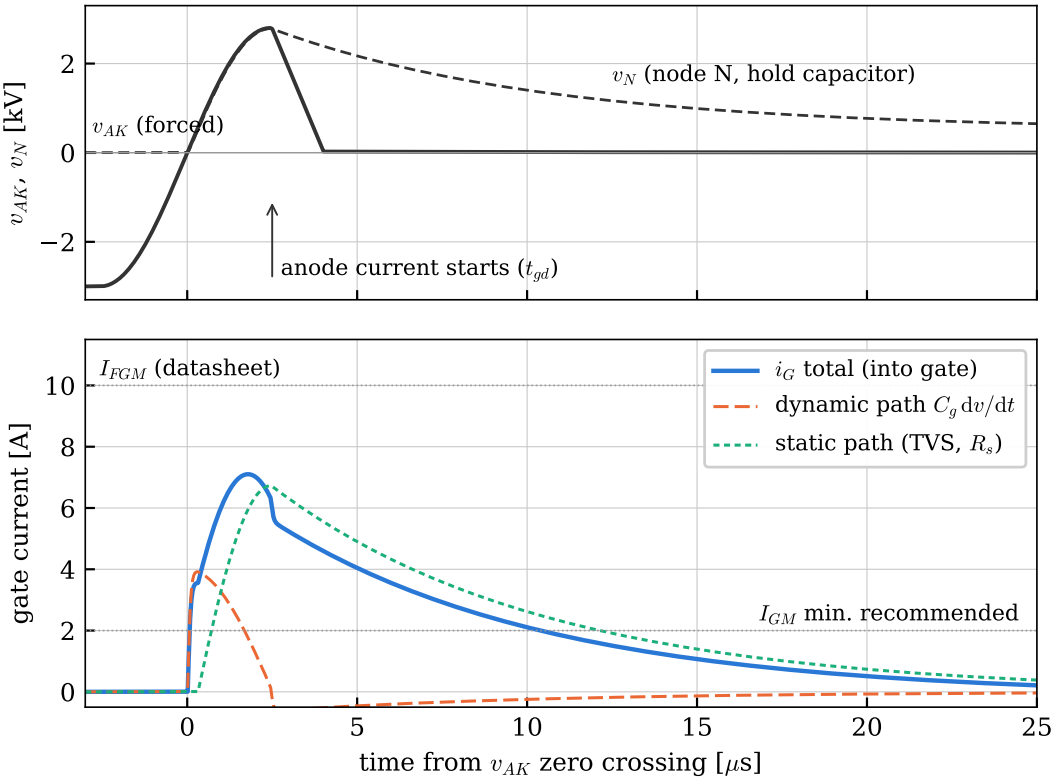


Figure 5: Computed response of the gate network to the breaker opening (nominal case, Table 3). Top: forced anode voltage and voltage of node N. Bottom: gate current and its two components; the static path, fed by C_h, sustains the pulse after the anode voltage has collapsed.

The nominal case gives a total gate current above the trigger level 30 ns after the zero crossing, above 5 A from 0.7 µs, a peak of 7.1 A at 1.8 µs ($v_{GK} \approx 6$ V), above 2 A until 10.4 µs and above 0.6 A until 18.3 µs; the gate charge is 54 µC. Table 4 summarises the sensitivity to the edge rate, to the MOV tolerance and to the presence of C_h: the limit I_{FGM} is reached only for an edge of about 4 kV/µs, outside the design space (Sec. 3.4); without C_h the pulse ends with the collapse of the anode voltage, 2.6 µs after its start, and fails the 5–20 µs recommendation.

Table 4: Sensitivity of the computed gate pulse (network of Table 3).

Case	$\hat{\imath}_G$	$\hat{\imath}_{Cg}$	$\hat{\imath}_{D1}$	$i_G > 2$ A	$i_G > 0.6$ A
Nominal, 1.8 kV/µs, plateau 2.5 kV	7.1 A	3.9 A	43 A	10.3 µs	18.3 µs
Fast edge, 3.0 kV/µs	8.5 A	6.4 A	72 A	9.7 µs	17.6 µs
Very fast edge, 4.5 kV/µs	10.9 A	9.3 A	106 A	9.5 µs	17.5 µs
MOV −10 %, plateau 2.0 kV	5.6 A	3.5 A	39 A	8.2 µs	16.1 µs
Nominal without C_h	7.5 A	4.2 A	17 A	2.6 µs	2.8 µs

7.6 Slow reversal of the load voltage

If the interruption upstream is not the IGBT breaker but a slow device – a mechanical breaker or a fuse – the load voltage reverses only when the arc voltage exceeds the source voltage, $u_{\text{load}} = V_s - v_{\text{arc}}$, and it does so at a rate of volts per microsecond or less. The dynamic path is then below its 300 V/µs threshold and the static path fires the crowbar at $v_{AK} = V_{\text{fire}} \approx 0.75$ kV, i.e. at $v_{\text{arc}} \approx 3.75$ kV. The commutation into L_c is then driven by V_{fire} only: $\mathrm{d}i/\mathrm{d}t \approx 60$ A/µs, $t_{\text{comm}} \approx 60$ µs, during which the arc voltage holds v_{AK} near V_{fire} and the static path keeps supplying 0.7–2 A; C_h is charged to V_{fire} only and adds little. A soft turn-on at 60 A/µs with 1 A of gate current is within the rating conditions. This case does not size the network; it only shows that the crowbar also protects the bench against an upstream interruption that is not the static switch, provided the interrupting device can generate the arc voltage.

7.7 Energy and dissipation per event

Per opening the network handles: $\frac{1}{2} C_h v_{pk}^2 \approx 86$ mJ (of which about 30 mJ in the TVS stack, 50 mJ in R_s, 3–5 mJ in R_C), $\frac{1}{2} C_g v_{pk}^2 \approx 9$ mJ (in R_{gs}, the gate and R_C), and the gate itself receives 54 µC at 3–6 V, about 0.3 mJ. The total is 0.1 J per event, i.e. 0.1 W average at one opening per second: component ratings are set by pulse voltage and pulse current, not by power.

7.8 Failure modes and supervision

The failure modes that lose the crowbar all show up in the load voltage: the plateau at −2.5 kV lasts about 20 µs when the crowbar works and about 2 ms when it does not (Fig. 2). A supervision rule in the bench controller – alarm and stop if $u_{\text{load}} < -1$ kV persists for more than 100 µs – protects the MOV bank against a silent loss of the crowbar. The gate-lead current transformer allows the pulse of Fig. 5 to be checked at commissioning and periodically.

7.9 Optional active trigger

If the breaker controller can issue a gate command, it should be used as the primary trigger (pulse transformer or fibre-optic gate unit) with the passive network of Fig. 4 left in place as an independent back-up – the same philosophy as HVDC valves. The command must be issued together with, not before, the IGBT turn-off: a positive gate pulse applied while the thyristor still blocks −3 kV produces the reverse-leakage amplification described in Sec. 6. With an active

Table 5: Failure modes of the network. "Crowbar lost" means that the MOV bank absorbs the full 16.6 kJ (Fig. 2); the thyristor itself is not endangered in any case.

Failure	Effect	Consequence / detection
D_1 open	No trigger at all.	Crowbar lost; detected by plateau supervision.
D_1 short	Network exposed to −3 kV in service: D_{GK} clamps the gate at −0.7 V; R_s (27 kW) and R_C (410 W) overloaded and open.	Thyristor safe; network destroyed, crowbar lost at the next event; 8 kV string rating makes the fault remote.
TVS open	Dynamic path only; no sustaining current after the collapse.	Firing probable but outside the gate recommendations; detect with the gate-lead current transformer.
TVS short	Static path becomes R_s from N to the gate: fires at $v_{AK} \approx 0.2$ kV.	Still functional; lower threshold.
C_h short	Node N shorted to K: both paths disabled.	Crowbar lost; detected by plateau supervision.
C_g short (one of two)	$C_g \to 4.7$ nF: 8.5 A at 1.8 kV/µs.	Still below I_{FGM}; this is why C_g is two capacitors in series.
R_{gs} open	Dynamic path lost; static path unaffected.	Functional.

unit the passive network is unchanged; the two pulses simply add at the gate, and the sum must still respect I_{FGM}.

8 Coordination with the MOV clamp

1. **Firing level:** $V_{\text{fire}} \ll V_{\text{MOV}} - V_s$. With $V_{\text{fire}} \approx 0.75$ kV and 2.0–2.8 kV available the margin is a factor 2.7–3.7; the dynamic path fires at the zero crossing.

2. **$\mathrm{d}i/\mathrm{d}t$:** $(V_{\text{MOV}} - V_s)/L_c = 200$–$224$ A/µs, independent of the trigger; equal to the 50 Hz rating, five times below the 1 Hz rating, with a gate pulse three times the rating conditions. The two knobs are L_c and the MOV clamp level (Secs. 3.2 and 3.3).

3. **MOV duty:** about 180 J per opening during the 20 µs of gate delay and commutation; the varistor bank becomes a back-up clamp with negligible ageing.

4. **MOV tolerance:** the network works from a plateau of 2.0 kV upwards (Table 4); a lower clamp level, if ever chosen to reduce the $\mathrm{d}i/\mathrm{d}t$, requires re-checking the sustaining current or lowering V_{st}.

5. **Branch snubber:** at firing C_{sn} discharges through R_{sn} into the thyristor, $v_{AK}/R_{sn} \leq 44$ A and $\frac{1}{2}C_{sn}v_{AK}^2 \leq 8$ J in R_{sn}: harmless.

9 Verification in simulation

With the existing Simscape model of the 5STP 26N6500 and the gate network of Table 3 added:

- Plot v_{AK} (device terminals, not branch voltage), i_G, v_{GK}, i_{D1} and i_A at breaker opening. Confirm that i_G exceeds I_{GT} while v_{AK} is still low, that the peak of i_G stays below 10 A,

that $i_G > 2\,\mathrm{A}$ for at least 5 µs after the anode voltage has collapsed, and record the forward voltage at which anode current starts. Compare with Fig. 5.

- Confirm the load-voltage waveform of Sec. 2.2: plateau of about 20 µs, then $\approx -1.5\,\mathrm{kV}$ decaying with 1.64 ms; MOV energy $\approx 180\,\mathrm{J}$.
- Extract $\mathrm{d}i_A/\mathrm{d}t$ during the first 20 µs and compare with the repetitive rating; evaluate $L_c = 15$–$18\,\mu\mathrm{H}$ if margin is wanted on the 50 Hz figure.
- Sweep the breaker edge rate from $0.3\,\mathrm{kV/\mu s}$ to $3\,\mathrm{kV/\mu s}$ (breaker snubber capacitance, IGBT gate resistance) and the MOV clamp level by ±10 %; the static path should take over below the threshold (14) and the sustaining current should remain above 4 A on the plateau.
- Breaker closing: v_{AK} steps from 0 to $-3\,\mathrm{kV}$; check that the gate stays quiet (v_{GK} within $\pm 1\,\mathrm{V}$), that node N does not move by more than a few volts, and that D_1 sees no overvoltage beyond its rating including the closing overshoot.
- Abnormal case: breaker re-closing, or reconnection of the low-voltage pre-heating source, while the crowbar still conducts. The source then forces the crowbar current to zero through L_c ($240\,\mathrm{A/\mu s}$ at 3 kV, $4\,\mathrm{A/\mu s}$ at 50 V) with reverse recovery; check the reverse voltage overshoot across T and the branch snubber. The turn-off time $t_q = 800\,\mu\mathrm{s}$ is irrelevant in normal operation, since forward voltage returns only at the next cycle.
- Slow opening (Sec. 7.6): ramp v_{AK} at $10\,\mathrm{V/\mu s}$ and confirm firing at $\approx V_{\mathrm{fire}}$ with the anode current rising at $60\,\mathrm{A/\mu s}$.

10 Conclusions

The crowbar thyristor of the 3 kV / 3600 A bench dissipates about 10 J per event and is sized by voltage, $\mathrm{d}i/\mathrm{d}t$ and cycle count. A symmetric 6.5 kV phase-control thyristor such as the 5STP 26N6500 is the appropriate choice: it blocks the 3 kV reverse DC of service life with adequate cosmic-ray margin, and its $\mathrm{d}i/\mathrm{d}t$ capability matches the 200–224 A/µs imposed by L_c and the MOV clamp. That $\mathrm{d}i/\mathrm{d}t$ is set by the dump inductor and not by the load inductance: during the transfer of the current from the varistor to the thyristor the 2 mH act as a constant current source, and the transfer rate is $(V_{\mathrm{MOV}} - V_s)/(L_c + L_\sigma)$ – without L_c it would be the kiloampere-per-microsecond figure of the bus-bar stray inductance. L_c is the trade-off between the thyristor $\mathrm{d}i/\mathrm{d}t$ and the MOV energy per event.

The simulated load voltage at breaker opening – a swing from +3 kV to −2.8 kV in 4–5 µs, a plateau at −2.5 kV and, without crowbar, 2 ms of MOV conduction with 16.6 kJ – shows that the intrinsic self-firing mechanisms cannot be relied upon: forward breakover is unreachable and the $\mathrm{d}v/\mathrm{d}t$ of 1.5–2 kV/µs is below the critical 3 kV/µs, so the device would simply block. A negative load voltage is, in this DC circuit with unidirectional current, an unambiguous signal that the path from the source has been interrupted, and the trigger network is designed to respond to it and to nothing else: the blocking diode D_1 keeps the network inert for $u_{\mathrm{load}} > 0$, the coupling capacitor fires the gate at the zero crossing of the load voltage, the TVS stack with its series resistor fires it at $u_{\mathrm{load}} \approx -0.75\,\mathrm{kV}$ on any slow reversal and sustains 5–7 A on the MOV plateau, and the hold capacitor extends the pulse to 10–18 µs after the anode voltage has collapsed, in line with the gate-drive recommendations of the manufacturer. The design values of Table 3 respect I_{FGM} and V_{RGM} with margin, handle about 0.1 J per event, and fail either safe or detectably; a supervision of the plateau duration in the bench controller and a current transformer on the gate lead complete the scheme. Firing level and $\mathrm{d}i/\mathrm{d}t$ are coordinated with the MOV clamp through $V_{\mathrm{fire}} \ll V_{\mathrm{MOV}} - V_s$ and $(V_{\mathrm{MOV}} - V_s)/L_c$.

References

[1] Hitachi Energy (formerly ABB Semiconductors), "Phase Control Thyristor 5STP 26N6500," data sheet.

[2] ABB Switzerland Ltd, Semiconductors, "Gate-drive recommendations for phase control thyristors," Application Note 5SYA 2034.

[3] B. J. Baliga, *Fundamentals of Power Semiconductor Devices*. New York: Springer, 2008, ch. 8 (Thyristors).

[4] "Protective circuitry for a thyristor," U.S. Patent 4 651 251 (break-over diode protective firing, Nelson River HVDC precedent).

[5] J. Arrillaga, *High Voltage Direct Current Transmission*, 2nd ed. London: IEE, 1998 (thyristor valves, protective firing).

[6] Dynex Semiconductor, "Pulsed power thyristors," product information, accessed Sept. 2026.

[7] Littelfuse / Vishay, "5KP series – 5000 W transient voltage suppressors," data sheet.

[8] ABB Switzerland Ltd, Semiconductors, "Design of RC snubbers for phase control applications," Application Note 5SYA 2020.